

Soheun Yi

M (878) 348-2113
E-mail: soheuny@andrew.cmu.edu
Website: soheunyi.github.io

Education

Carnegie Mellon University, *Ph.D., Statistics.*

Aug 2023–May 2028 (*Expected*)

Carnegie Mellon University, *M.S., Statistics.*

Aug 2023–May 2025

Seoul National University, *B.S., Mathematical Sciences.*

Mar 2017–Aug 2023

◦ *Summa Cum Laude*

◦ On leave from Fall 2019 to Spring 2021 for mandatory military service in the Republic of Korea.

Research Interests

- Statistical theory and methodology for **dynamical systems** and **optimal transport**.
- Statistics and machine learning for **scientific discovery**.

Research Experience

Graduate Research Assistant, Carnegie Mellon University.

2023–Present

Advisors: Sivaraman Balakrishnan and Gonzalo Mena.

- Developed estimators for velocity fields from repeated marginal samples, connecting distributional dynamics estimation to optimal transport; established finite-sample upper bounds and matching minimax lower bounds [7].
- Introduced Concept Modulation Models and algebraic criteria for identifiability and extrapolation in conditional latent-variable models, connecting nonlinear ICA and causal representation learning [1].
- Developed a model-agnostic new-physics search using density-ratio learning to locate localized high-frequency signal regions; evaluated it on simulated LHC $HH \rightarrow 4b$ events [6, 8].

Researcher, Krafton, Deep Learning Division.

Apr–Jul 2023

- Studied mini-batch contrastive learning and developed a spectral-clustering approach to identify high-loss batches and accelerate training relative to standard batching [4].
- Researched neural radiance field models for 3D scene reconstruction.

Research Intern, Causality Lab, Seoul National University.

2022–2023

Advisor: Sanghack Lee.

- Developed Filter, Rank, and Prune for causal discovery in linear cyclic Gaussian graphical models [5].
- Designed a BIC-inspired score balancing fit and sparsity and evaluated the method against cyclic-graph baselines.

Research Intern, Ryu Optimization Group, Seoul National University.

2022

Advisor: Ernest K. Ryu.

- Proved tighter convergence guarantees for Davis–Yin splitting using scaled relative graphs [2].
- Extended the analysis to convex optimization problems, establishing improved convergence rates [3].

Publications

(* denotes equal contribution.)

- [1] **Soheun Yi**, Y. Lu, C. Squires, and P. Ravikumar. “Concept Modulation Models: A Unified Framework for Identifiability and Extrapolation”. *Advances in Neural Information Processing Systems (NeurIPS)*. 2026.
- [2] J. Lee*, **Soheun Yi***, and E. K. Ryu. “Convergence Analyses of Davis–Yin Splitting via Scaled Relative Graphs”. *SIAM Journal on Optimization* (2025).
- [3] **Soheun Yi** and E. K. Ryu. “Convergence analyses of Davis–Yin splitting via scaled relative graphs II: convex optimization problems”. *Optimization* (2025).
- [4] J. Cho, K. Sreenivasan, K. Lee, K. Mun, **Soheun Yi**, J. Lee, A. Lee, J. Sohn, D. Papailiopoulos, and K. Lee. “Mini-Batch Optimization of Contrastive Loss”. *Transactions on Machine Learning Research* (2024).
- [5] **Soheun Yi** and S. Lee. “Filter, Rank, and Prune: Learning Linear Cyclic Gaussian Graphical Models”. *Proceedings of The 27th International Conference on Artificial Intelligence and Statistics (AISTATS)*. 2024.
- [6] **Soheun Yi**, J. Alison, and M. Kuusela. “Toward Model-Agnostic Detection of New Physics Using Data-Driven Signal Regions”. *ML4PS Workshop at NeurIPS*. 2024.

Preprints

- [7] **Soheun Yi**, G. Mena, and S. Balakrishnan. "Minimax Velocity Field Estimation from Snapshots and Trajectories". In preparation. 2026.
- [8] **Soheun Yi**, J. Alison, and M. Kuusela. "Model-Agnostic Search for New Physics in the Presence of Imperfect Background by Smearing the Data". In preparation. 2026.

Industry Experience

Quantitative Developer, *Hyperithm*, Seoul.

2019–2021

- Researched, backtested, and deployed quantitative trading strategies for digital-asset markets.
- Built tools to monitor live strategies and processed market and transaction data using Python and BigQuery.

Talks

2026: "Minimax Velocity Field Estimation," contributed paper, JSM.

2024: "Toward Model-Agnostic Detection of New Physics Using Data-Driven Signal Regions," ML4PS Workshop at NeurIPS.

Awards and Honors

2023–Present: Overseas Ph.D. Scholarship, Korea Foundation for Advanced Studies.

2020: Pair Merit Prize, Simon Marais Mathematics Competition (top 4 of 150 pairs).

2019–2023: Undergraduate Scholarship, Korea Foundation for Advanced Studies.

2018: Gold Prize, Korea Undergraduates Mathematics Competition.

2017–2023: Presidential Science Scholarship, Korea Student Aid Foundation.

2016: Finalist for Korean IMO Team Selection; Silver Medal, Romanian Master of Mathematics.

Teaching and Service

CMU: Teaching assistant: Deep Learning Theory (2026), Advanced Statistical Theory (2025), Intermediate Statistics (2024–2026), Advanced Data Analysis (2024), and Modern Regression (2023).

SNU: Teaching assistant: Mathematical and Numerical Optimization (2022).

Other: Organizer, CMU Optimal Transport Reading Group (2025–2026); Deputy Leader, Team Korea at Romanian Master of Mathematics (2019); Korea Mathematics Olympiad Winter School teaching assistant (2018).

Skills

Methods: Optimal transport, distributional modeling, minimax theory, statistical inference, convex optimization, causal discovery, and density-ratio estimation.

Tools: Python, SciPy, NumPy, Pandas, PyTorch, CVXPY, Git, and Linux/HPC (SLURM).